to be second, the Soviets were turning hardware originally intended for manned lunar flight to another purpose. Their Soyuz spacecraft carried out a series of maneuvers and dockings in space, with at one time three of the craft in orbit at once. In June 1970 the crew of Soyuz 9 set a new record of 18 days in orbit. This was the prelude to the launch of the first space station, Salyut 1, on April 19, 1971.

Troubled debut

Salyut 1 was designed to accommodate three cosmonauts, who would have a range of scientific and observation instruments to keep them usefully employed on board. As it turned out, no one was ever to recount their experience of this space novelty. The first crew sent up, on Soyuz 10, docked with Salyut 1 but could not get inside it. The next crew, on Soyuz 11, spent three weeks on board Salyut 1 but died during their return to Earth after depressurization of the descent module.

This tragedy was followed by other setbacks for the Salyut program which meant that it was not until June 1974 that a Soviet crew successfully traveled to an orbiting station (Salyut 3), spent some time on board, and returned to Earth. By then, the United States had carried out its own first space station experiment.

Looking beyond the Moon landings, leaders of the US manned space program had always seen space stations as a part of their future plans, along with a reusable space vehicle and the more distant goal of bases on the Moon and Mars. The barrier was cost. Although NASA presented space stations and a reusable space vehicle as interdependent parts of the same package, budgetary constraints dictated that it had to choose between them. As a result, Skylab, launched on May 14, 1973, was a one-time, a relatively cheap orbiting station using hardware adapted from the Moon flights. Despite the undoubted success of Skylab, there was no follow-on to a sustained US space station program.

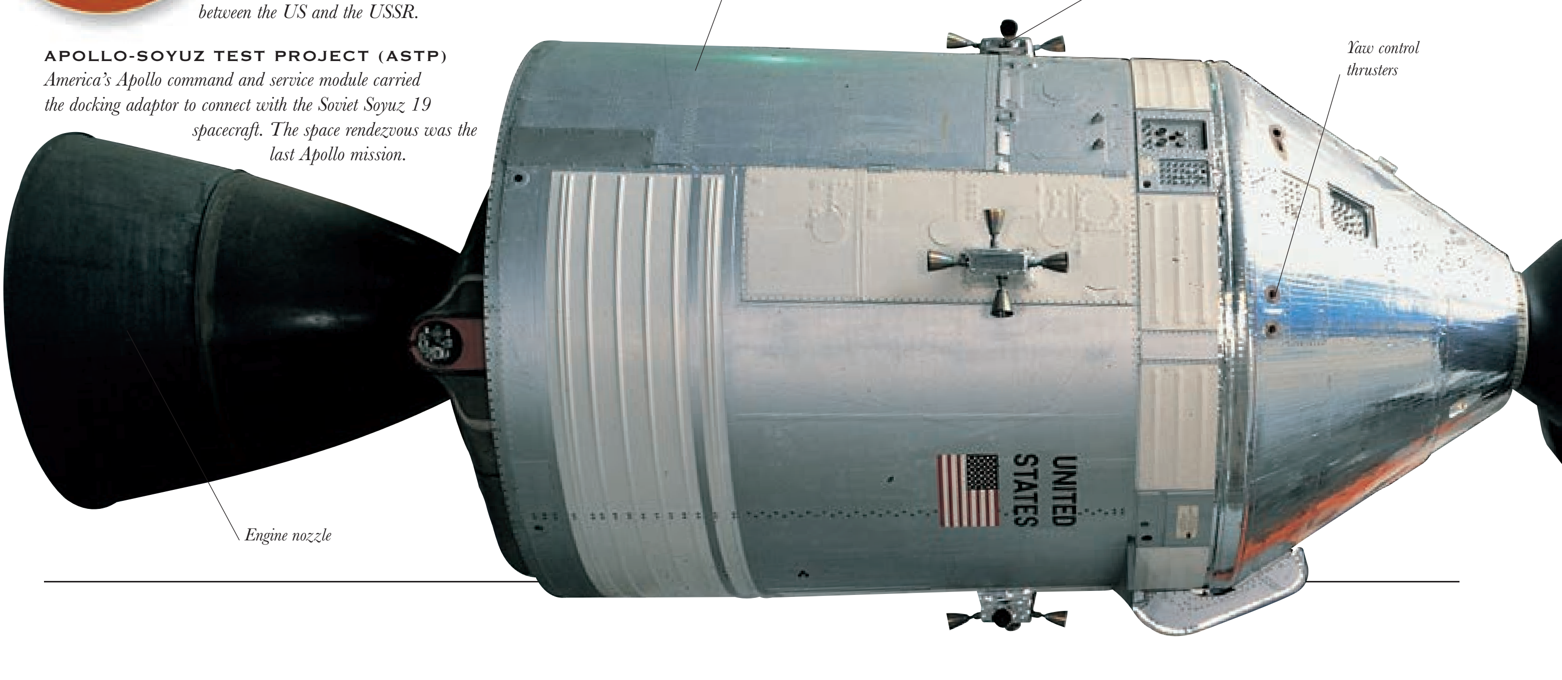
The last US astronauts left Skylab in February 1974. Only one American crew went into space in the next seven years. They were on board an Apollo launched for the Apollo-Soyuz Test Project, a highly publicized symbolic gesture of international cooperation between Cold War adversaries engaged in a policy of detente. The Apollo and Soyuz spacecraft rendezvoused in space on July 17, 1975. Soviet and American spacemen exchanged gifts and friendly phrases, spent two days working together, and went back to Earth.

From then until the spring of 1981, Soviet cosmonauts had space to themselves. A succession of military and civilian Salyuts were sent up into orbit in a program that stubbornly overcame difficulties and setbacks. The early Salyuts were crude, uncomfortable, and shortlived. Even the most successful, Salyut 4, which was inhabited by two crews for sustained periods, was reportedly cold and damp. However, Salyut 6, launched in September 1977, marked a significant step forward. Refueled by unmanned space tankers, it was able repeatedly to use its rocket engine to counter orbital decay, staying in orbit until 1982. Its successor, Salyut 7, lasted from 1982 to 1991. Ferried to these space stations by rapidly improving Soyuz craft, cosmonauts broke space endurance records time and again – Valery Ryumin and Leonid Popov stayed aloft for 185

> "All the inexperienced cosmonauts agitate for longer space flights, but the veterans of long-term flights don't."
>
> **Oleg Gazenko**
> Director of the Institute for Biomedical Problems

DOOMED CREW
Vladislav Volkov, Viktor Patsayev, and Georgi Dobrovolsky, the crew of Soyuz 11, became the first men to die outside the Earth's atmosphere when the air leaked from their descent module.

APOLLO-SOYUZ LINKUP
These badges celebrate the Apollo-Soyuz space rendezvous in 1975, an unprecedented gesture of cooperation between the US and the USSR.



APOLLO-SOYUZ TEST PROJECT (ASTP)
America's Apollo command and service module carried the docking adaptor to connect with the Soviet Soyuz 19 spacecraft. The space rendezvous was the last Apollo mission.